1.6c

Sensation
Hearing

Audition helps us survive

The sounds we hear best are those in the range of the human voice.

A fraction of second after a stimulus, millions of neurons simultaneously coordinated extracting essential features, comparing with past & identifying stimulus

Sound Waves

amplitude (height) → perceived loudness
frequency (measured hz) → pitch

long waves have low freq & low pitch

Sound intensity measured in decibals.

0 decibals = absolute threshold for hearing
+10 decibals → 10x increase in sound intensity

Prolonged exposure to sounds >85 db → hearing loss

The Ear

Sound waves strike eardrum → tight membrane vibrates.

Outer ear funnels sound to eardrum.

Middle ear: Hammer (malleus), Anvil (incus), Stirrup (stapes)

Bones of middle ear form a piston to amplify & relay eardrum's vibrations and transmits them to the Cochlea: snail-shaped tube in inner year.

Incoming vibrations cause the oval window, the Cochlea's membrane-covered opening, to vibrate, jostling fluid inside Cochlea.

This motion ripples basilar membrane, bending hair cells lining its surface.

Hair cell movements trigger impulses in adjacent neurons, where axons converge for auditory nerve.

Auditory nerve carries messages to thalamus then auditory cortex in temporal lobe.

Cochlea has 16k hair cells compared to 130m photoreceptors in eye.

but hair cells are very responsive.

Deflect cilia on its tip by an atom's width

Δ it will detect.

Damage to hair cell receptors or auditory nerve can cause sensorineural hearing loss. (nerve deafness)

Conduction hearing loss, damage to the mechanical system, is less common.

A cochlear implant, a kind of bionic ear, can restore hearing.

Loudness

Brain interprets loudness from number of activated hair cells.

Two theories on discriminating pitches.

- Place theory :: different waves trigger activity in different parts along basal membrane.
- Frequency theory :: brain monitors freq of nu impulses

If wave has freq 100 $\frac{\text{wave}}{\text{sec}}$, 100 pulses travel 1 second

But individual neurons can't fire >1000 /sec.

So volley theory proposes that some cells rest while others fire.

localization

Our placement of years gives us 3d hearing.

MCQ: 1d, 2b, 3a, 4c, 5a